

SIMATS ENGINEERING

OBJECT-ORIENTED ANALYSIS AND UML- BASED DESIGN

AI-Enabled Disaster Resilience and Emergency Response Platform (AIDERP)

Course: CSA1104 - Object Oriented Analysis and Design

Code of Conduct

I N. Hima Harshitha (192371063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the academic-integrity requirements of the assessment. I understand that any violation of academic-integrity rules may result in disciplinary action.

Signature of the student: N. Hima Harshitha

1. Problem Analysis and Requirement Summary

AIDERP is a resilient, role-based emergency coordination platform for floods, cyclones, fires, earthquakes, heatwaves and other disaster events. It combines citizen reports, IoT sensor readings, weather and geospatial feeds, hospital and shelter capacity, emergency-service updates, and resource data. The platform helps responders act on verified information while ensuring that AI recommendations remain decision support under human oversight.

System Boundary

The system boundary includes incident intake, validation, risk prioritisation, emergency coordination, alerts, safe-route guidance, capacity management, resource tracking, authentication, audit logging, and integration adapters. Physical rescue work, final medical triage, weather forecasting, telecom delivery, and external GIS routing are external services or human responsibilities.

Primary Stakeholder Needs

Stakeholder	Need
Citizens	Report danger, share location, request assistance, receive verified alerts and safe routes.
Emergency operators	Validate incidents, prioritise cases, dispatch teams, monitor workload and issue alerts.
Responders	Receive assignments, navigate safely, update status and request resources.
Hospitals / shelters	Report occupancy, beds, supplies and service availability.
Administrators	Manage users, roles, policies, reports and audit evidence.

2. Functional and Non-Functional Requirements

Functional Requirements

ID	Requirement
FR-01	Citizen can create an incident report with type, description, location, media and assistance request.
FR-02	System accepts IoT readings and creates a preliminary incident when configured thresholds are breached.
FR-03	Operator can verify, reject, merge, prioritise and escalate incidents.
FR-04	AI risk engine calculates a transparent priority recommendation using severity, exposure, accessibility, sensors and resource availability.
FR-05	Operator can dispatch responders and resources and track assignment status.
FR-06	Citizen can view verified alerts, nearest safe shelter and recommended route.
FR-07	Hospital and shelter staff can update capacity, beds, supplies and availability.
FR-08	System sends role-appropriate notifications through configured channels.
FR-09	Administrator manages users, roles, service zones and configuration.
FR-10	System records audit events for security-sensitive actions.

Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Availability	Critical reporting and alert functions remain available during partial network failure.
NFR-02	Security	Use authenticated sessions, RBAC, encrypted transport and protected audit logs.
NFR-03	Privacy	Collect minimum location/person data; expose it only to authorised roles.
NFR-04	Performance	Create urgent incident acknowledgement within 3 seconds under normal load.
NFR-05	Scalability	Allow horizontal API/notification scaling during surges.

NFR-06	Maintainability	Use cohesive modules, stable interfaces and documented status vocabulary.
NFR-07	Resilience	Queue messages and support graceful degradation when external feeds fail.

3. Actor Identification

Actor	Responsibilities
Citizen	Reports incidents, requests help, receives alerts and routes.
Emergency Operator	Verifies incidents, overrides AI recommendation, dispatches and monitors operations.
Field Responder	Accepts assignments, reports observations, requests resources and resolves work.
Hospital Staff	Updates hospital service capacity and emergency intake availability.
Shelter Manager	Updates beds, occupancy, supplies and operating status.
Volunteer / Utility Team	Receives approved assignments and reports task progress.
Administrator	Manages roles, policy, configuration, audit and reports.
IoT Sensor / Gateway	Publishes sensor readings and threshold alerts.
External Services	Weather, GIS/routing, map, SMS/email/push and identity providers.

4. Use Case Descriptions

Use Case	Primary Actor	Description
UC-01: Report Incident	Citizen	Citizen supplies incident type, location, evidence and help request. System validates data, creates IncidentReport, acknowledges receipt and places it in verification queue. Exception: unavailable network stores encrypted local draft and retries.
UC-02: Assess and Prioritise	AI Risk Engine / Operator	System combines verified evidence, severity, population exposure, access, sensor readings and capacity. It returns a score, reasons and recommended resources. Operator can accept or override with reason.
UC-03: Verify and Dispatch	Emergency Operator	Operator reviews evidence, marks verified/rejected/merged, selects responders/resources and publishes assignments and alerts.
UC-04: Update Response	Field Responder	Responder accepts assignment, moves through En Route, On Scene and Resolved, submits observations and resource requests.
UC-05: Evacuation Guidance	Citizen	Citizen opens verified alert; system returns recommended safe route and available shelters.
UC-06: Update Capacity	Hospital Staff / Shelter Manager	Authorised user updates occupancy, bed count, supplies and availability; system validates and publishes aggregate availability.
UC-07: Sensor Incident	IoT Sensor	Gateway submits signed reading. If threshold and confidence rules pass, system creates a preliminary incident for human verification.

5. Use Case Diagram

Use Case Diagram - AIDERP System Boundary

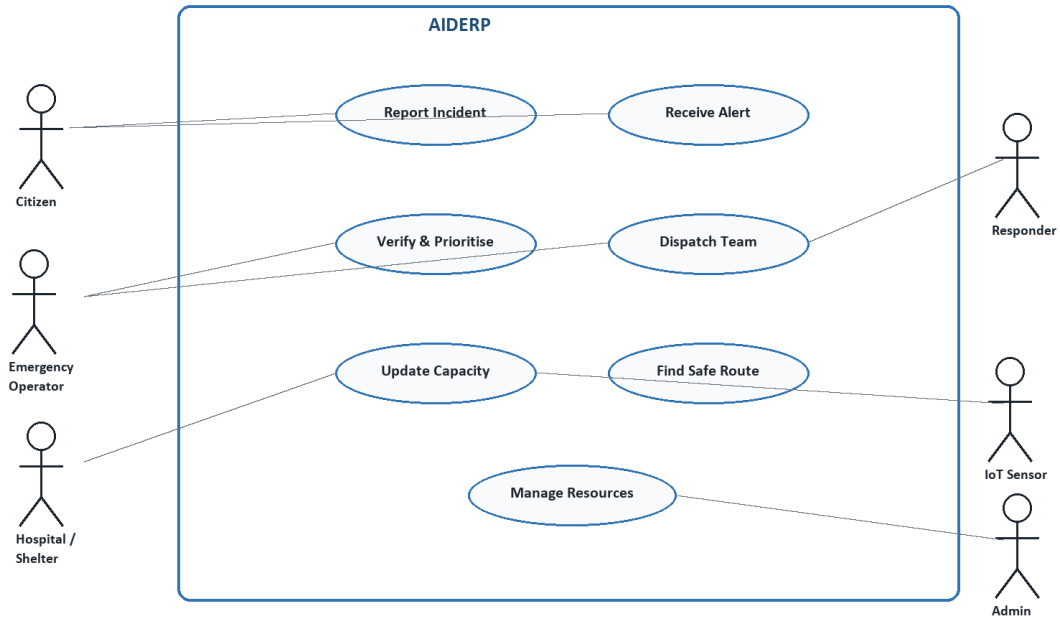


Figure 1. AIDERP use case diagram. System functions are available only to the relevant actor roles.

6. Object and Class Identification

Class	Key Attributes	Key Operations
User	userId, name, role, contact	authenticate(), receiveAlert()
Incident	incidentId, type, severity, status, location	verify(), prioritise(), updateStatus()
IncidentReport	reportId, evidence, submittedAt, verified	submit(), validate()
RiskAssessment	score, priority, reasons	assess(), recommendResources()
ResponseAssignment	assignmentId, responder, status	assign(), accept(), complete()
Resource	resourceId, type, availability	reserve(), release()
Shelter / Hospital	capacity, occupancy, location	updateCapacity(), reportAvailability()

SensorReading	sensorId, type, value, timestamp	validate(), createPreliminaryIncident()
EmergencyAlert	alertId, message, area, level	publish(), expire()
AuditLog	eventId, actorId, action, timestamp	record(), search()

7. Class Diagram

Class Diagram - Core Domain Model

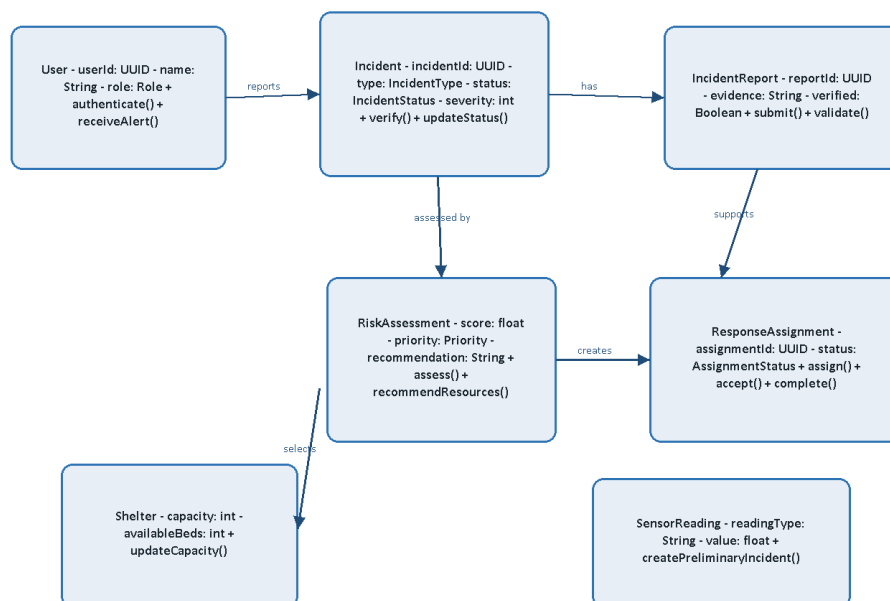


Figure 2. Core class diagram. Incident is the central aggregate; reports, assessments and assignments have explicit responsibilities.

8. Sequence / Interaction Diagrams

The following interaction models cover the assignment's key scenarios. The same message contracts are reused in the implementation: report submission, AI assessment, operator verification, dispatch, responder update, route lookup, capacity update and sensor ingestion.

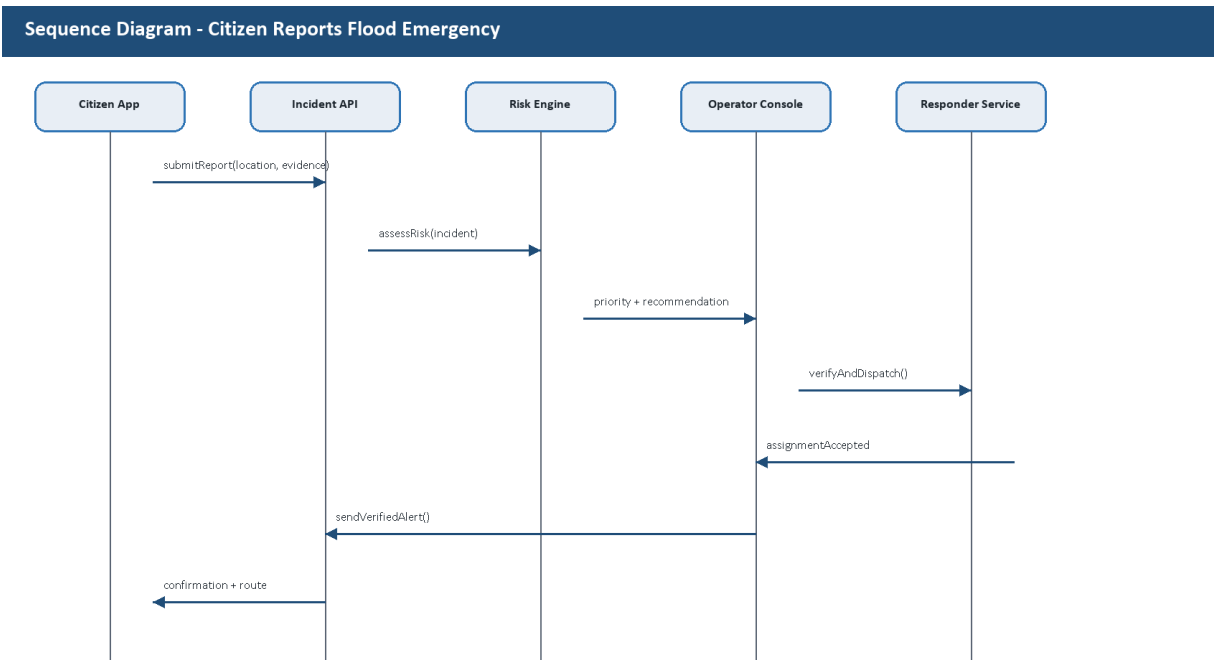


Figure 3. Sequence diagram for a citizen flood report, AI prioritisation, operator verification and responder dispatch.

Interaction summaries for remaining scenarios

Scenario	Interaction Flow
Responder status update	Responder App -> Assignment API: accept/update status -> Incident Service: validate transition -> Notification Service: notify operator/citizen -> AuditLog: record action.
Evacuation route	Citizen App -> Alert API: request guidance -> GIS Adapter: request safe route -> Shelter Service: rank available shelters -> Citizen App: display route and capacity.
Capacity update	Hospital/Shelter Portal -> Capacity Service: submit availability -> validate authorised role -> persist -> Dashboard/Operator Console receives update.
IoT threshold event	Sensor Gateway -> IoT Adapter: signed reading -> threshold rule -> create preliminary Incident -> Operator Console notified for verification.

9. Activity Diagram

Activity Diagram - Emergency Response Workflow

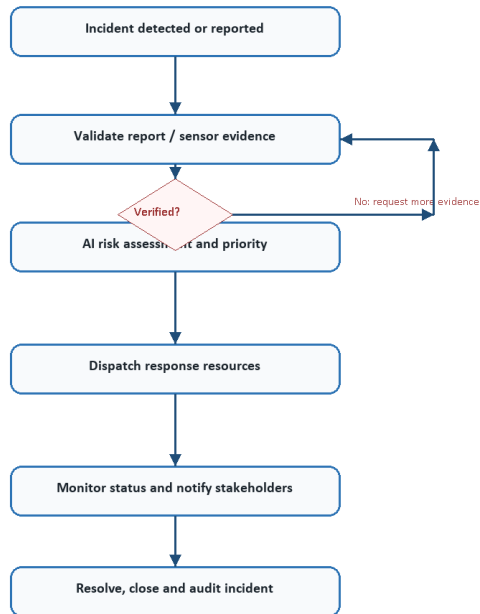


Figure 4. Activity diagram for the emergency response workflow, including validation and feedback when evidence is insufficient.

10. State Transition Diagram

State Transition Diagram - Incident Lifecycle

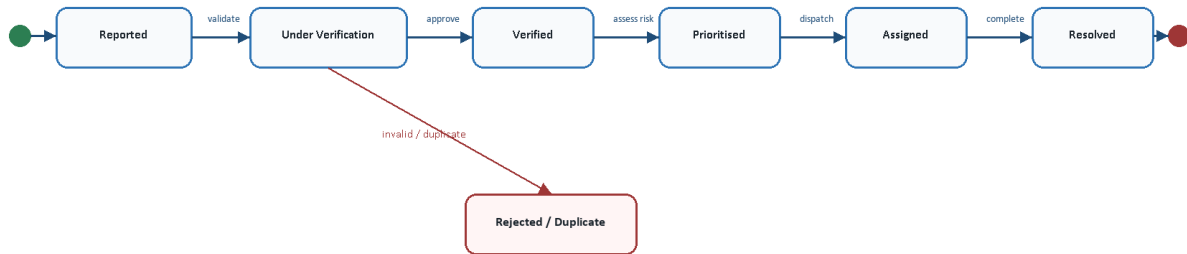


Figure 5. Incident state transition diagram. Only authorised transitions are permitted and each transition is audited.

11. Package Diagram

Package Diagram - Logical Modules

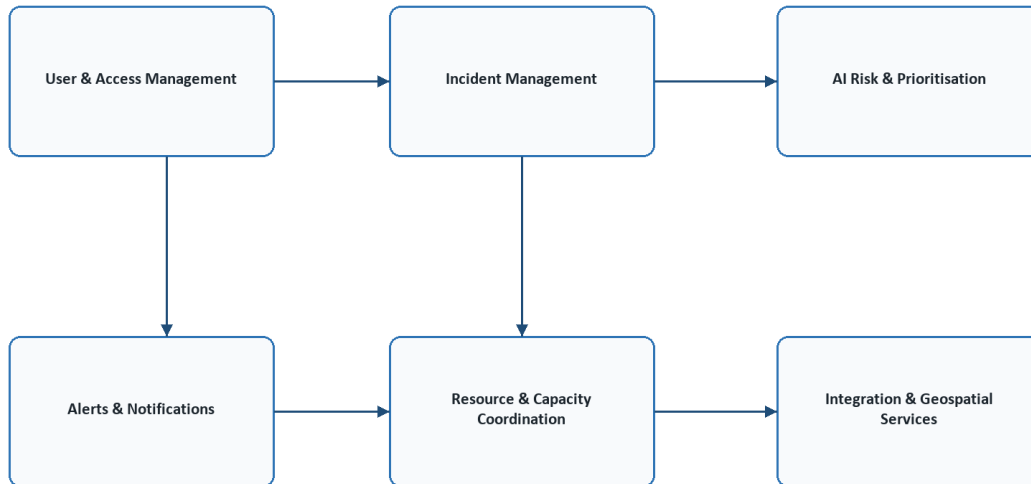


Figure 6. Package diagram showing logical separation between access, incident, AI, notification, capacity and external-integration concerns.

12. Component Diagram

Component Diagram - Service Architecture

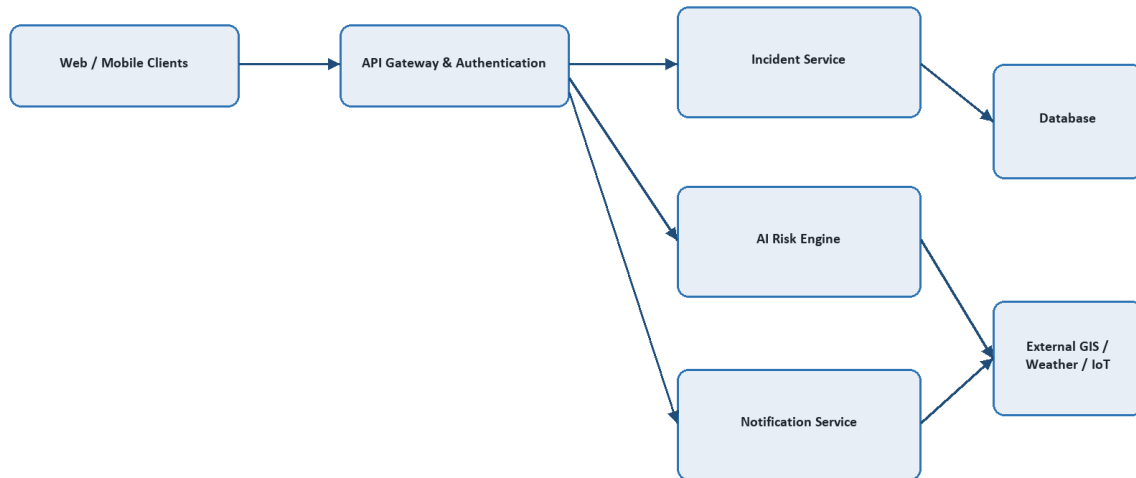


Figure 7. Component diagram. Authentication and authorisation occur before services handle incident or resource data.

13. Deployment Diagram

Deployment Diagram - AIDERP Runtime Topology

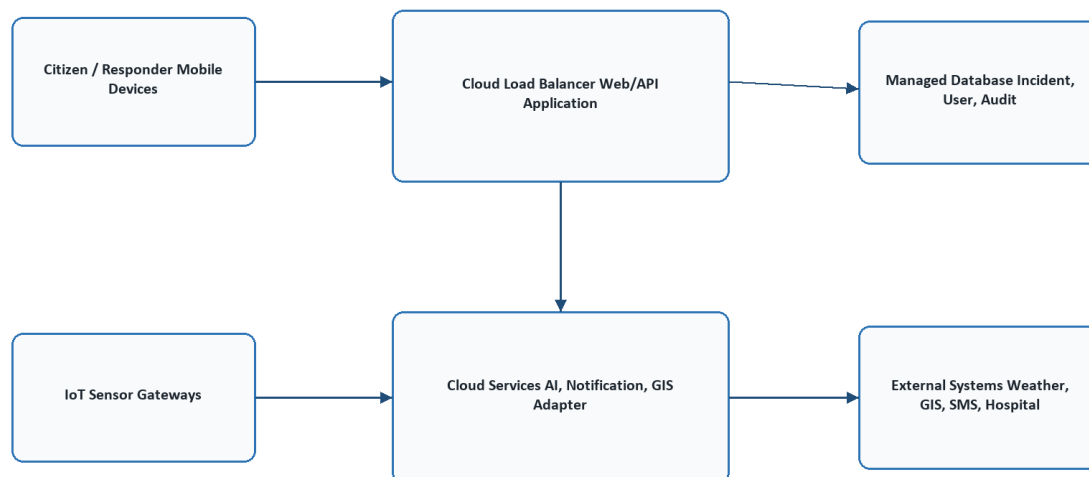


Figure 8. Deployment diagram. Mobile clients, IoT gateways, cloud services, database and external systems are deployed as separate nodes.

14. Object Responsibility Analysis

Object	Responsibility	Collaborators
Incident	Owns lifecycle and authoritative incident status.	IncidentReport, RiskAssessment, ResponseAssignment
RiskAssessment	Produces explainable recommendation; does not make final command decision.	Incident, SensorReading, Resource, Operator
ResponseAssignment	Tracks responder/resource allocation and completion.	Incident, User, Resource
Shelter/Hospital	Owns capacity and availability facts for its facility.	CapacityService, Operator
NotificationService	Delivers role-aware messages through channels.	EmergencyAlert, User, external gateways

AuditLog	Records security-sensitive state changes and access actions.	All service-layer operations
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Responsibility assignment follows the Information Expert principle: the object that owns the relevant data owns the behaviour. Controllers coordinate request flow but do not contain business calculation logic. This prevents a single “God class” and maintains high cohesion.

15. Design Axioms, Principles and Design Pattern Justification

Principle / Pattern	Application and Justification
Encapsulation	Incident status changes only through validated methods; internal data is not directly mutated by client screens.
Abstraction	Interfaces hide weather, GIS, notification and IoT supplier differences.
High cohesion / low coupling	Each package has a narrow purpose; services communicate through explicit DTO/API contracts.
SOLID	Single responsibility for services; open extension through adapters; interface segregation for external APIs; dependency inversion through service interfaces.
State Pattern	Incident lifecycle state controls allowed transition rules.
Strategy Pattern	Risk calculation can change policy/configuration without changing incident workflow.
Observer Pattern	Verified alerts and assignment updates notify subscribed users/channels.
Factory Pattern	Incident creation selects correct incident subtype from citizen, operator or sensor source.
Adapter Pattern	Adapters normalise weather, GIS and IoT provider APIs.

16. Assumptions, Constraints, Security and Privacy Considerations

Area	Content
Assumptions	Users have registered identities; external map/weather and notification providers are available through approved APIs; operators retain final control over dispatch decisions.

Constraints	Network outages, device battery limits, incomplete sensor readings, high event volume, inaccessible roads and uncertain evidence may degrade data quality.
Verification	Citizen and sensor reports are preliminary until an authorised operator verifies, rejects or merges them.
Security	TLS, signed API requests where appropriate, RBAC, rate limits, audit logging, input validation, protected secrets and secure session controls are required.
Privacy	Collect minimum necessary personal/location data; restrict exact location and medical/capacity detail by role; encrypt sensitive records; define retention and deletion policy.
AI ethics	Display recommendation reasons/confidence; allow human override; monitor bias; never issue an irreversible command solely from AI output.

17. Requirement-to-Use Case/Class Traceability Matrix

Requirement	Use Case	Supporting Classes
FR-01	UC-01	IncidentReport, Incident, Location
FR-02	UC-07	SensorReading, Incident, IoTAdapter
FR-03	UC-03	Incident, IncidentReport, Operator
FR-04	UC-02	RiskAssessment, Resource, SensorReading
FR-05	UC-03 / UC-04	ResponseAssignment, Resource, Responder
FR-06	UC-05	EmergencyAlert, SafeRoute, Shelter
FR-07	UC-06	Shelter, Hospital, CapacityService
FR-08	UC-03 / UC-05	NotificationService, EmergencyAlert
FR-09	Administration	User, Role, AuditLog
FR-10	All privileged use cases	AuditLog, SecurityService

18. Implementation / Prototype and Results

A suitable prototype is a responsive web application with separate Citizen, Operator, Responder, Facility and Administrator views. The minimum demonstrable workflow is: a citizen submits a flood incident; the system creates a pending Incident; the risk engine returns severity and resource recommendation; an operator verifies and dispatches a responder; the responder updates En Route, On Scene and Resolved; the citizen receives a verified alert and shelter guidance. A facility update and sensor-created preliminary incident should also be demonstrated.

Test Case	Expected Result	Illustrative Prototype Result
T-01: Citizen report	Report is validated, geo-tagged and placed in verification queue.	Pass - valid report acknowledged; missing location rejected.
T-02: Operator verification	Operator can verify/merge/reject and audit event is created.	Pass - verified report receives priority recommendation.
T-03: Dispatch	Eligible responder receives assignment and can accept.	Pass - status moved to Assigned then En Route.
T-04: Capacity update	Facility capacity changes are validated and visible to operator.	Pass - full shelter excluded from recommendation.
T-05: IoT threshold	Threshold breach creates preliminary incident, not auto-verified incident.	Pass - operator review required.
T-06: Access control	Unauthorised user cannot dispatch or view restricted location data.	Pass - API returns access denied.

The reported results are prototype verification outcomes on controlled example data. They demonstrate model consistency and system behaviour; they are not evidence of real-world disaster-response effectiveness.

19. Reflection on Design Decisions, SDG Relevance and Learning Outcomes

The key design decision was to separate unverified input from verified incidents and to treat AI output as explainable decision support. This avoids a dangerous assumption that automated scoring is always correct. The State and Strategy patterns made the incident lifecycle and risk policy easier to extend. The

package/component separation also reduces coupling between core emergency workflow and volatile external providers.

Area	Reflection
SDG 3 - Good Health and Well-being	Supports faster coordination with hospitals, medical teams and safe shelters.
SDG 9 - Industry, Innovation and Infrastructure	Uses interoperable service architecture, IoT integration and resilient digital infrastructure.
SDG 11 - Sustainable Cities and Communities	Improves community warnings, shelter awareness and coordinated incident management.
SDG 13 - Climate Action	Supports preparedness and response to climate-related floods, cyclones, heatwaves and fires.
Learning outcomes	Applied OO analysis, actor/use case modelling, object classification, UML structural/behavioural/architectural diagrams, traceability, design patterns and quality attributes.

20. GitHub Upload

Create a GitHub repository named `aiderp-ooad-assignment`. Upload the final report, editable diagram source files, prototype source code, screenshots, test cases and a README. Do not upload API keys, passwords, personal emergency reports or production credentials.

Recommended repository structure

```
aiderp-ooad-assignment/  
  README.md  
  report/AIDERP_OOAD_Complete_Assignment.docx  
  diagrams/use-case.png  
  diagrams/class.png  
  diagrams/sequence.png  
  prototype/  
  tests/test-cases.md  
  docs/assumptions-and-traceability.md
```

README checklist

- Project purpose and stakeholder summary
- Setup/run instructions for the prototype

- Diagram inventory and requirement traceability
- Test cases and known limitations
- Team/student contribution and academic-integrity statement

References

Object Management Group, Unified Modeling Language (UML) specification.

ISO/IEC 25010, Systems and software Quality Requirements and Evaluation (SQuaRE).

United Nations, Sustainable Development Goals 3, 9, 11 and 13.